

Vision Classification and Retrieval-Augmented Generation for Automatic Waste Management

Course assignment report: Vision Classification + RAG

Repository: <https://github.com/eshanaajith/vision-rag-waste>

Five-class waste image classification with grounded handling guidance

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1. Introduction

This project combines a convolutional neural network (CNN) with retrieval-augmented generation (RAG) for household waste. Given an image and a natural-language question, the system predicts one of five classes—Glass, Leather, Organic, Paper, or Plastic—then retrieves passages from official waste-management documents and returns a short, cited answer. The design constraint is grounding: the answer must be taken from indexed source text. A language model is not used to invent handling procedures. Qwen 2.5 was tested separately and is not on the answer path.

2. System Design

2.1 Image classifier

WasteCNN is a three-block CNN. RGB images are resized to 128×128 and normalised with ImageNet mean and standard deviation. Each block applies a 3×3 convolution (32, then 64, then 128 channels), ReLU, and 2×2 max-pooling, producing a 128×16×16 feature map. A fully connected head maps this to 128 units (ReLU, dropout 0.5) and then to five logits. Training uses cross-entropy loss, Adam (learning rate 0.001), batch size 32, and 20 epochs. The checkpoint with the best validation accuracy is saved as *best_waste_cnn.pth*. Training augmentation is horizontal flip and ±10° rotation. Inference supports JPEG, PNG, and HEIC.

2.2 Knowledge base and retrieval

Five public documents form the knowledge base (Table 1). Text is split into overlapping chunks of 1,000 characters with 200-character overlap, yielding 559 chunks, most from S1. Chunks are embedded with *sentence-transformers/all-MiniLM-L6-v2* and stored in a FAISS inner-product index (normalised embeddings, equivalent to cosine similarity). At query time the system retrieves 30 candidates, then keeps the five highest-scoring chunks that mention the predicted waste class (or an alias such as cardboard, PET, or rexine).

Table 1. Knowledge-base sources.

ID	Title	Origin
S1	Solid Waste Management Rules, 2026	India Gazette
S2	Guidelines for Citizens	Maharashtra Pollution Control Board
S3	How Do I Recycle Common Recyclables	U.S. EPA
S4	Composting	U.S. EPA
S5	Recycling Basics and Benefits	U.S. EPA

2.3 Extractive answering

Retrieved chunks are split into sentences and scored for (i) explicit class mention, (ii) handling or recycling actions, (iii) question intent (handle, recycle, or compost), and (iv) instructional wording. Background statistics, shopping advice, and mixed dry-waste lists are penalised. At most two sentences are returned, cited only with the source IDs that contributed those sentences. If no sentence exceeds the relevance threshold, the system returns: “The available sources do not provide sufficient information to answer this question.” For sparse classes such as leather, a short context window is allowed: if a nearby sentence or the same chunk names the class, a handling rule in that chunk may be used, with a one-sentence lead-in that the class is listed among recyclable waste in the guidelines.

3. Dataset and Experimental Setup

The dataset contains 1,973 labelled photographs across the five classes. Images were split 70% / 15% / 15% into train, validation, and test sets with stratified sampling and random seed 42 (Table 2). Leather is the smallest class. No image

appears in more than one split. Classifier metrics are reported on the held-out test set of 296 images. End-to-end RAG is evaluated on 15 test images (three per class) with a class-specific handling or recycling question.

Table 2. Class distribution by split.

	Glass	Leather	Organic	Paper	Plastic	Total
Train (70%)	290	204	283	317	287	1,381
Validation (15%)	63	44	60	68	61	296
Test (15%)	62	43	61	68	62	296
All	415	291	404	453	410	1,973

4. Results

4.1 Classification

On the test set the CNN reaches 97.30% accuracy and macro-averaged F1 of 0.974 (Table 3). Plastic is classified without error. Organic, leather, and glass have recall 1.00. Paper is the weakest class (recall 0.8824): eight paper images are misclassified, six of them as glass. Glass precision is therefore 0.9118. The confusion matrix is given in Table 4. Per-image predictions and softmax confidence scores are stored in *results/test_predictions.csv*.

Table 3. Test-set precision, recall, and F1 (296 images).

Class	Precision	Recall	F1	Support
Glass	0.9118	1.0000	0.9538	62
Leather	0.9773	1.0000	0.9885	43
Organic	0.9839	1.0000	0.9919	61
Paper	1.0000	0.8824	0.9375	68
Plastic	1.0000	1.0000	1.0000	62
Macro average	0.9746	0.9765	0.9743	296
Accuracy	0.9730 (97.30%)			296

Table 4. Test-set confusion matrix (rows: true class; columns: predicted class).

	Glass	Leather	Organic	Paper	Plastic
Glass	62	0	0	0	0
Leather	0	43	0	0	0
Organic	0	0	61	0	0
Paper	6	1	1	60	0
Plastic	0	0	0	0	62

4.2 End-to-end evaluation

Fifteen test images (three per class) were classified and then queried with a handling or recycling question. All 15 images were classified correctly, typically with confidence above 0.99. Answers were identical across the three images of each class. Table 5 summarises one representative case per class. The full 15-row table, including image paths, confidence, retrieved source scores, answers, and support status, is submitted as *results/end_to_end_evaluation.csv*. Every selected answer sentence is taken from a retrieved chunk and cited; all 15 cases are marked Supported.

Table 5. Representative end-to-end cases (three images per class; answers identical within class).

Class	Question	Retrieved	Answer (abbrev.) and citation
Glass	How should glass waste be handled?	S2, S3, S1	Broken glass should not go into the recycling bin. [S3]
Leather	How should leather waste be handled?	S2	Leather is listed among recyclable waste; recyclable waste can be sold to a raddiwala or scrap dealer. [S2]
Organic	How should organic waste be handled?	S4, S2	Dig a compost pit and put biodegradable waste into it for manure. [S2]
Paper	How should paper waste be handled?	S2, S3, S1	Most community or office recycling programs accept paper and paper products. [S3]
Plastic	How should plastic waste be recycled?	S2, S3	Plastic is listed among recyclable waste; give it to collection agencies; if segregated at source, it can be recycled. [S2]

5. Observed Failure Cases and Limitations

Paper–glass confusion. All eight test errors are paper images. Six are labelled glass, often with high confidence (for example 0.999 on *20241228_151006.jpg* and *20241209_140414.jpg*). Likely causes are visual overlap (white or translucent sheets, glare) and 128×128 resolution, which removes texture that would distinguish paper from glass. One paper image is predicted as leather with low confidence (0.46 on *20250129_124751.jpg*); another is predicted as organic (0.91 on *20250228_153739.jpg*). Because glass recall is perfect, these errors reduce glass precision rather than glass recall. A larger input size or a pretrained backbone would be the natural next step; this assignment used a small CNN trained from scratch.

Sparse leather evidence. S2 lists Leather and Rexine in a recyclable-waste table and, in a later sentence of the same chunk, states that recyclable waste may be sold to a raddiwala. No source gives a leather-only procedure. The system therefore links the class mention to the nearby handling rule. That is conservative relative to inventing a method, but it is not a same-sentence leather instruction. If the window were disabled, leather questions would return insufficient information.

Source mix and jurisdiction. Paper answers currently prefer the EPA recycling-program sentence [S3] over the MPCB dry-waste handover sentence [S2], even though S2 is retrieved. Glass guidance (broken glass and the recycling bin) is likewise EPA text. The index mixes Indian rules (S1, S2) with U.S. EPA pages (S3–S5). The pipeline does not rank by jurisdiction, so a locally correct practice may be omitted if a foreign sentence scores higher.

Extractive style. Answers cannot synthesise a complete local procedure when the corpus only contains a fragment. That is intentional: the system refuses to hallucinate. The cost is that answers can be short or slightly generic (the plastic answer uses a class lead-in plus a recyclable-waste collection rule). Retrieval quality is also bounded by chunking: S1 dominates the 559 chunks, so FAISS can surface regulatory text that is on-topic by embedding similarity but weak as a household instruction.

End-to-end sample size. The 15-case table does not measure RAG quality on misclassified images. If the CNN labelled paper as glass, the retriever would search glass evidence and the answer would be wrong even if the text were well grounded. Classifier errors therefore propagate. The 15 selected images were all classified correctly, so this failure mode is not visible in that table.

6. Conclusion

The submitted system is a working vision-plus-RAG pipeline: a five-class CNN (97.30% test accuracy, macro F1 0.974) followed by FAISS retrieval over five cited sources and extractive, class-aware answering. On 15 held-out images the classifier is correct in every case and each answer is supported by a retrieved sentence. Remaining weaknesses are concentrated in paper–glass visual confusion, sparse leather text, and mixed Indian/U.S. guidance. Those limits follow from the dataset, the small CNN, and the decision not to generate ungrounded advice.